Chapter 6

Recovering Data

IT 360 | Undelete | Damaged Media

🧹 Why Recover Data?

- Users delete data — file or **remnants** may still be available to the examiner
- Deleting is trivially easy; non-savvy criminals think it hides evidence
- Expect evidence to be deleted on many computers you examine
- Retrieving deleted data = one of the most **fundamental** forensic tasks

Topics

- How to undelete data
- Recovering information from damaged drives

🎯 Goals

- Recover deleted files in **Windows**
- Recover deleted files in **Linux**
- Recover deleted files in **Mac OS**
- Recover files from **damaged drives**

💿 File Systems & Hard Drives

Sectors and Clusters

Layer	Views data as
Hard drive	Sectors – **512 bytes**, or **4,096 bytes** (Advanced Format)
File system / OS	**Clusters** – 1 to 128 sectors

- Sector = area of a platter defined by two radii
- Cluster sectors need **not** be contiguous
- Formal name changed to **allocation unit** – "cluster" still common

Clusters Are All-or-Nothing

File system treats a cluster as **fully used** if even one bit is used

- Example: 4,096-byte sectors × 10 sectors = **40,960-byte** cluster
- Save a **42,000-byte** file → needs **2 clusters**
  - Cluster 1: all 10 sectors used
  - Cluster 2: only 1 sector used — but FS & OS see it as **completely used**

Windows

FAT16 / FAT32

File Allocation Table maps every cluster. Each entry records one of:

1. Next cluster number for this file
2. **End of cluster chain** (EOC)
3. **Bad** cluster
4. **Reserved** cluster
5. **Open / available** cluster

Deleting on FAT

- Data **not removed** — FAT just marks clusters as no longer in use
- New saves may **overwrite** those clusters
- More recent the deletion → more likely to recover
- Recovery is not all-or-nothing → **partial** recovery possible

NTFS (Windows 2000 → today)

Two key files for forensics:

- **MFT** (Master / Meta File Table)
  - One base record per file & directory
  - Filenames, time stamps, security IDs, attributes (read-only, compressed, encrypted…)
  - Same role as the FAT
- **Cluster bitmap**
  - One bit per cluster → allocated or free

Deleting on NTFS

1. File marked **"deleted"** → moved to **Recycle Bin** (no bits removed)
2. User empties Recycle Bin → clusters marked available
3. MFT filename updated with a **special character**
4. Data still there → **entire file** recoverable until overwritten

Windows	Recycle Bin folder
Pre-Vista	`RECYCLER` (hidden)
Vista +	`$Recycle.Bin` (hidden)

Recycle Bin – Windows 2000 / XP

- Folder `Recycler\%SID%` – one per user (SID = Security Identifier)
- Hidden **INFO2** file per user tracks:
  - Original location, file size, deletion time
- → Ties deleted files to **specific users**

```
D%DriveLetter%%IndexNumber%.%FileExtension%
```

- **IndexNumber** = order of deletion · no extension = folder

Recycle Bin – Windows 7 → 10

INFO2 is gone. `$Recycle.Bin\%SID%`:

- `$R` `xxxxxx.ext` → the original file (renamed)
- `$I` `xxxxxx.ext` → metadata, always **544 bytes**

`$I` bytes	Contents
0–7	Header: `01` + seven `00`
8–15	Original file size (hex, little-endian)
16–23	Deletion time – seconds since **Jan 1, 1601**
24–543	Original path / filename

Windows Recovery Tools

Tool	Notes
DiskDigger	Free; one file at a time (paid = bulk)
WinUndelete	3-step wizard: drive → file types → output folder
FreeUndelete	Free for personal use; select drive → Scan
OSForensics	"Deleted Files Search" – image or live system
Autopsy	Free, open source; recovers deleted files on image load
EnCase / FTK	Commercial suites – built-in recovery

DiskDigger: Dig Deep vs. Dig Deeper

Dig Deep	Dig Deeper
Scans **file system**	Scans **entire disk surface** (file carving)
Recovers file names	No file names
All file types	Only certain file types
Skips free space	Independent of file system/partitions

Result colors: 🟢 full file · ⚪ partial · 🔴 very little left

Autopsy

- Free, open-source forensics tool
- Starts recovering deleted files **as soon as an image loads**
- Lacks some advanced commercial features
- Excellent **second tool** to validate findings
- Ideal for students to experiment

Forensically Scrubbing Data

- Common claim: DoD 5220.22-M says overwrite **7 times**
- Accurate but incomplete → DoD has a **matrix** per media type
- Methods include:
  - **Degauss** (Type I / Type II degausser)
  - **Overwrite** – single char; or char, complement, random + verify
  - **Full chip erase** / UV erase (EPROM types)
  - **Remove all power** (volatile RAM)
  - **Destroy** – disintegrate, incinerate, pulverize, shred, melt



Linux

Ext File System

- Most common Linux FS – **Ext3** & **Ext4**
- Data stored in contiguous **blocks** (1,024 / 2,048 / 4,096 bytes)
- Block size kept in the **superblock**
- Blocks grouped; each group has:
  - Block **bitmap** (allocated blocks) – up to 32,768 blocks/group
  - **Inode** bitmap + inode table

Inodes and Links

Inode = 128-byte structure holding everything about a file **except name & data**

Link	Behavior
Hard link	Points directly to file; OS counts references → count hits **0** = file deleted
Soft / symbolic link	Pointer to another file/dir — like a Windows shortcut

Manual Recovery with `grep`

1. Warn users with `wall` → switch to single-user mode: `init 1`
2. Search raw partition for known text:

```
grep -i -a -B10 -A100 'forensics' /dev/sda2 > file.txt
```

Flag	Meaning
`-i`	Ignore case
`-a`	Treat binary as text
`-B` / `-A`	Context before / after match

Linux Run Levels (`init`)

Run level	Description
1	Single-user mode
2	Not used (user-definable)
3	Full multiuser, no GUI
4	Not used (user-definable)
5	Full multiuser with GUI

> “Unfamiliar with an OS? Leave its forensics to someone qualified — but learn Linux fundamentals.”

Linux Tools

extundelete – Ext3 & Ext4

```
extundelete /dev/sda4 --restore-all
```

Scalpel – file carver (Linux, Mac; compilable for Windows)

```
sudo apt-get install scalpel
# uncomment wanted types in /etc/scalpel/scalpel.conf
sudo scalpel [device/dir/file] -o [output dir]
```

- Output directory must be **empty**

🍎 Mac OS

Mac OS Recovery

- Since OS X, based on **FreeBSD** (UNIX clone) → many Linux techniques work
- File systems:
  - **HFS+** (Hierarchical File System Plus) – replaced HFS
  - **APFS** (Apple File System) – replaced HFS+ in **High Sierra (2017)**
- **MacKeeper** – free trial; Files Recovery → select volume → **Undelete**
- Be comfortable with the OS before analyzing it

🔧 Damaged Media

Physical Damage

- CDs: scratched substrate/dye · HDDs: head crash, failed motor · Tapes: break
- Physical damage → data loss, often **logical damage** too
- End users usually **can't** repair — attempts make it worse
- Only orgs with special gear (**clean rooms**) should repair

> “**Clean room** – environment with controlled levels of dust, microbes, particles”

Evaluating a "Failed" Hard Drive

1. Connect drive to a working **test system** (data cable only)
2. Boot test system from its own drive — **listen**: platters spinning?
3. Recognized? → copy all files; virus-check; verify integrity
4. Not spinning / not recognized → limited repair, then image (e.g., **DCFLdd**)
5. Still failing → send to **data recovery specialists**

Last-Resort Repairs

If data is already considered "lost":

- Swap the **circuit board** from a matching healthy drive
- Swap the **read/write head** assembly
- Move the **platters** into a healthy drive

Logical Damage

More common than physical damage

- **Causes:** power outages, shutdown during boot/shutdown, RAID controller/driver errors, crashes
- **Symptoms:** crashes, data loss, recursive directories, negative free space
- **Repair tools:** `chkdsk` (Windows) · `fsck` (Linux) · Disk Utility (Mac) · Sleuth Kit · **TestDisk**
- **Journaling** FS (NTFS 5.0, Ext3) roll back to a stable state

Logical Recovery Techniques

	Consistency Checking	Zero-Knowledge Analysis
How	Verify structure vs. spec (e.g., `.` & `..` entries)	Rebuild FS from scratch using known structure
Speed	Fast	Much slower
Weakness	Fails if badly damaged; `chkdsk` may **delete** files	Doesn't repair – extracts data elsewhere
Tools	`chkdsk`, `fsck`	Scan for structures & file boundaries

File Carving

- Extract a single file from a larger data set (disk/partition)
- Used when metadata damaged or file corrupt
- Finds **headers / footers**, pulls data between them
- Tools: **Scalpel**, **carver-recovery**

File	Header (hex)	File	Header (hex)
JPEG	`FF D8`	PDF	`25 50`
BMP	`42 4D`	ZIP	`50 4B`
EXE	`4D 5A`	PNG	`89 50`
GIF	`47 49`	WAV / AVI	`52 49`
MP3	`49 44`		

📝 Summary

- Deletion rarely removes data — FS tables just mark space free
- Know the FS: **FAT / NTFS**, **Ext**, **HFS+ / APFS**
- Recycle Bin artifacts (`INFO2` , `$I` / `$R`) tie deletions to users
- Use multiple tools; validate with a second tool (e.g., Autopsy)
- Damaged media: assess carefully, image first, carve when needed